Anatomy of a Migration

Migrations are subclasses of the Rails class `ActiveRecord::Migration`. When necessary, migrations can contain `up` and `down` methods:

```
class SomeMeaningfulName < ActiveRecord::Migration
  def up
    # ...
  end

  def down
    # ...
  end
end
```

The name of the class, after all uppercase letters are downcased and preceded by an underscore, must match the portion of the filename after the version number. For example, the previous class could be found in a file named `20221207000017_some_meaningful_name.rb`. No two migrations can contain classes with the same name.

The `up` method is responsible for applying the schema changes for this migration, while the `down` method undoes those changes. Let's make this more concrete. Here's a migration that adds an `e_mail` column to the `orders` table:

```
class AddEmailToOrders < ActiveRecord::Migration
  def up
    add_column :orders, :e_mail, :string
  end

  def down
    remove_column :orders, :e_mail
  end
end
```

See how the `down` method undoes the effect of the `up` method? You can also see a bit of duplication here. In many cases, Rails can detect how to automatically undo a given operation. For example, the opposite of `add_column` is clearly `remove_column`. In such cases, by simply renaming `up` to `change`, you can eliminate